

Test On Programming (Test 1)

(Solve any 5)

Marks: 50 (5 x10)

****Write Variable Description Table for each program****

1. Write a program to take three numbers and find second smallest
2. Write a program to take a number and check if it's leap year, century leap year, or just century year
3. Write a program to take units consumed and calculate electricity bill based on the following chart

Units	Rate
First 100 units	Rs. 7.5
Next 200 units	Rs. 9
More than 300 units	Rs. 10

4. Write a program to take consumed data from user and print internet bill

Data in GB	Rate per 10 GB
50 GB	Rs.10
Next 30 GB	Rs.8
Next 20 GB	Rs. 6
More than 100	Rs. 5

5. Write a menu driven program to convert Celsius to Fahrenheit or Fahrenheit to Celsius
Based on user's choice

Celsius-Fahrenheit Relation: $\frac{c}{5} = \frac{f-32}{9}$

6. Take a, b and c from user, and find roots if discriminant = $\sqrt{b^2 - 4ac}$ ≥ 0 using the quadratic formula

Quadratic Formula : $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Nature of roots:

If **discriminant** > 0 : Two real and distinct roots.

If **discriminant** $= 0$: Two real and equal roots.

If **discriminant** < 0 : Two complex (imaginary) roots.

For now, we are not concerned about the complex or imaginary roots

7. Write a program to take a 5-digit number, find sum of first, last and middle digit. Check if the sum is even or odd